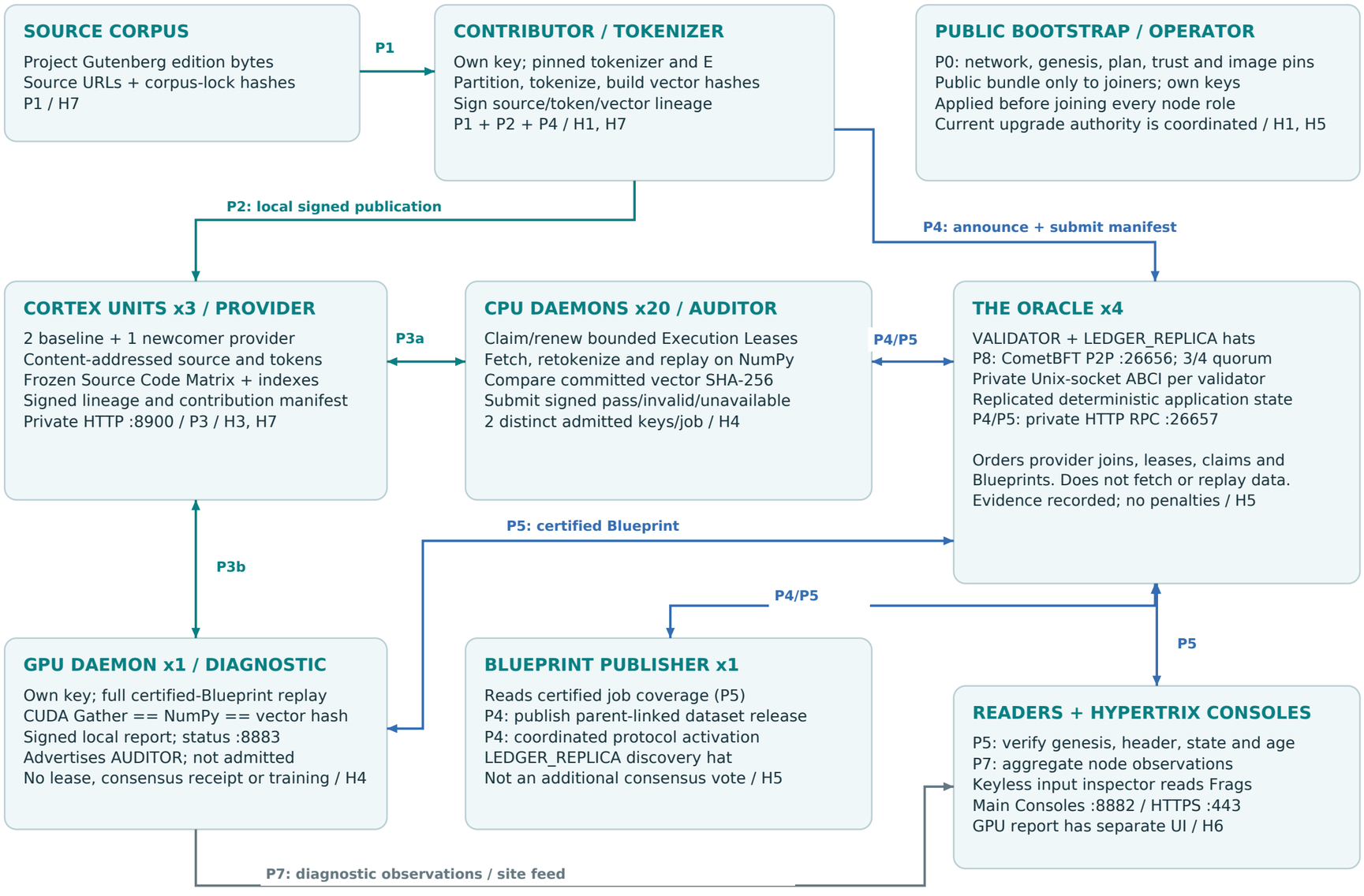
Node roles, artifact flow and ledger decisions

Logical responsibilities are shown separately; the newcomer combines ingestion/tokenization and one of the three Cortex providers in one container.

artifact / local publication · signed command / certified state · observation (not consensus)

P0-P9: protocol inventory on the next sheet.



THE COLLECTIVE / P6 DISCOVERY ON EVERY NODE

Distributed Kademlia-style HTTP :8903. Signed role/endpoint records, XOR routing, expiry and sequence checks. Lookup does not authorize a role. H1/H2.

FUTURE PATH / NOT EXECUTED IN THIS PROTOTYPE

Certified Blueprint + parent weights -> training node -> checkpoint validation / evaluations -> weight-ledger promotion. H8.

H1 / Role admission and identity

P0/P4: scoped, expiring role credentials; revocation. CA-issued or network-approved certificates are alternatives. A key is not a unique operator.

H2 / Eclipse, Sybil and discovery DoS

P6: diverse independent seeds and peer selection; per-peer request/storage quotas. Signed records alone do not make routing Sybil-resistant.

H3 / Data availability under churn

P3: replicas on independent hosts; repair and unpredictable retrieval checks. Hashes detect bad bytes; they do not make a missing provider available.

H4 / Collusion and dummy approvals

P4: post-commit unpredictable challenges, independent reassignment, canaries and dispute review. Two signatures do not prove honest computation.

H5 / Consensus and upgrade authority

P8/P0: independent validator operators, protected remote signers with fencing, threshold-reviewed releases and recovery drills. No automatic slashing now.

H6 / Transport, host and telemetry

P9/P7: peer-bound TLS/Noise or managed mTLS; workload ACLs and isolation; quotas and signed telemetry where needed. Transport auth does not grant a role.

H7 / Poisoning and source governance

P1/P2: rights/consent policy, deduplication, quarantine and corpus-quality review. Provenance binds origin and bytes, not truth or suitability.

H8 / Future training and weights

Future: bind dataset, parent weights, recipe and environment; validate checkpoints and evaluations before promotion. Exact Gather is not training reproducibility.

Transport boundaries and protocol inventory

P0-P9 identify the actual paths on the role diagram. All amber controls remain candidates; they have not been enabled by this documentation change.

LOCAL HOST / ONE FAILURE DOMAIN

2 Oracle + 2 Cortex + 16 CPU Daemons
Publisher + GPU diagnostic + local Console
23 containers total; own node keys/state mounts
ABCI uses each validator's private Unix socket
Public bootstrap is read-only for new joiners.

P9 / AUTHENTICATED SSH

Control: P4/P5/P6/P8 + telemetry
Bulk artifacts: P3
Additional route: GPU P6 only

AZURE ACI / ONE FAILURE DOMAIN

2 Oracle + 1 Cortex + 4 CPU Daemons + gateway + Console
9 containers; shared host, networking and lifecycle
Gateway has no validator keys; restricted forwarding targets
Public :2222 -> SSH gateway; :80/:443 -> Caddy -> Console :8882
Private application/DHT ports are not public ingress.

CURRENT TRUST BOUNDARY: one operator controls both sites, bootstrap and upgrade activation. Containers separate state; they do not create independent operators.

ID	Interface	Sender / receiver / transport	Payload, checks and important limits
P0	Bootstrap / runtime trust	Public files -> every role; operator controls issuance	SHA-256 network/genesis/plan/image pins; Ed25519 registrations. Own private state per node. Runtime upgrade is publisher-signed, not threshold governance.
P1	Source acquisition	External corpus -> contributor; HTTPS	Original edition bytes + URL/lock. Content hashing records what was fetched; HTTPS and lineage do not establish factual quality or rights.
P2	Ingestion / local provenance	Contributor -> its Cortex object store; local calls/files	Pinned tokenizer; uint32-le tokens; E[t] float32-le; SHA-256 artifact descriptors; Ed25519 signed parent DAG and author sequence. Dense vectors discarded.
P3a / P3b	Artifact pull / data plane	Private HTTP :8900 -> requester caches	CPU Puller: 1-MiB pieces, deduplication, bounded concurrency, retry/failover, final whole-object hash. GPU/public reader: bounded whole objects. Frozen E is transferred once per empty cache.
P4	Signed commands / leases	Contributor, admitted CPU Daemons and publisher -> Oracle RPC :26657	Canonical JSON + Ed25519; chain/plan binding, actor sequence and validity height. Announce/submit, claim/renew/result and publish. GPU diagnostic does not send these transactions.
P5	Certified reads	Oracle RPC :26657 -> CPU/GPU nodes, publisher, readers and Consoles	Go/CometBFT verifies the fixed-genesis quorum header and state app hash. Minimum accepted height + age window guard reads. Blueprint manifests then bind artifact fetches.
P6	Role discovery / peer routing	Each peer <-> peers/seeds; HTTP :8903	160-bit XOR distance; k=8, alpha=3; signed endpoint/role records, sequence and expiry, lifetime <=180 s. Announcements refresh every 25 s. Invited loopback routes, not open Internet discovery.
P7	Observability / public inspection	Node JSON files/site HTTP -> Console; browser HTTPS :443	Private cloud feed :8901, local feed forwarded :8902; local Console :8882 and GPU page :8883. Certificates are checked separately from unsigned telemetry. Read-only UI; no runtime socket.
P8	Consensus / application execution	Oracle <-> Oracle on TCP :26656; private ABCI Unix sockets	CometBFT fixed voting set; 3 of 4 votes. Per-validator deterministic state and persistent signer state. Incident/conflicting-result evidence can be recorded; no automatic financial punishment.
P9	Cross-host pilot transport	Local outbound SSH -> Azure gateway TCP :2222	Pinned SSH host key and operator authentication; allowlisted forwards. Separate control and bulk-data connections plus GPU DHT route. HTTP is private inside this mesh; no mesh-wide peer mTLS yet.

ROLE SCOPE: VALIDATOR, PROVIDER, TOKENIZER, AUDITOR and LEDGER_REPLICA are advertised hats. Actual audit/vote authority comes from admission rules; the GPU only advertises AUDITOR. TRAINER_GPU is a planned execution capability, not a running training node.

Implementation references: client.py, state.py, incidents.py, worker.py, dht.py, transfer.py, contributor_node.py, gpu_reconstruction_node.py and hybrid_observer.py. Security rationale and validation criteria: docs/architecture_notes.md.